# Unified Automatic Rigging and 4D Gaussian Reconstruction in Robust Structure

by

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# ABSTRACT

We propose a unified 4D dynamic reconstruction framework that integrates Gaussian-based neural rendering with the **UniRig** universal **skeleton-driven** rigging system for editable and physically consistent motion generation from monocular videos. Starting from an initial static 3D Gaussian representation, the framework first extracts a mesh surface and automatically derives a robust hierarchical skeleton together with skinning weights using UniRig. This enables the decomposition of complex object dynamics into two complementary motion components: skeleton-driven rigid articulation and skinning-based non-rigid deformation. A 2D–3D closed-loop optimization mechanism further ensures temporal consistency and geometric fidelity via reconstruction loss between rendered projections and input frames.

**Keywords:** 4D reconstruction, Gaussian Splatting, UniRig, skeleton-driven rigging

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# 1. Introduction

## 1.1 Background

Automatic skeleton rigging is a fundamental yet labor-intensive step in the 3D animation pipeline. Traditional **skeletal binding** requires significant manual design and domain expertise, often taking hours per character. To reduce this cost, learning-based rigging methods have been widely studied and can be broadly categorized into template-based and template-free approaches.

Template-based methods, such as SMPL-based systems [1, 2, 3, 4], rely on predefined skeletal structures and perform well on constrained topology spaces, but lack generalization to unseen or irregular shapes. Template-free methods, including RigNet [5] and volumetric or graph-based approaches [6, 7, 8], infer skeletons directly from geometry, improving flexibility but suffering from unstable topology and weak motion transfer.

Mesh deformation-based methods [9, 10, 11] avoid explicit skeletons but depend heavily on training data and are difficult to integrate into standard animation pipelines. Overall, existing methods still struggle with complex and diverse 3D assets, especially in large-scale generative settings, and are further limited by the lack of comprehensive datasets for automatic binding.

In addition, real-world mesh distributions vary significantly: artist-created meshes are structured and joint-aware, whereas AI-generated meshes often contain irregular and semantically inconsistent topology.

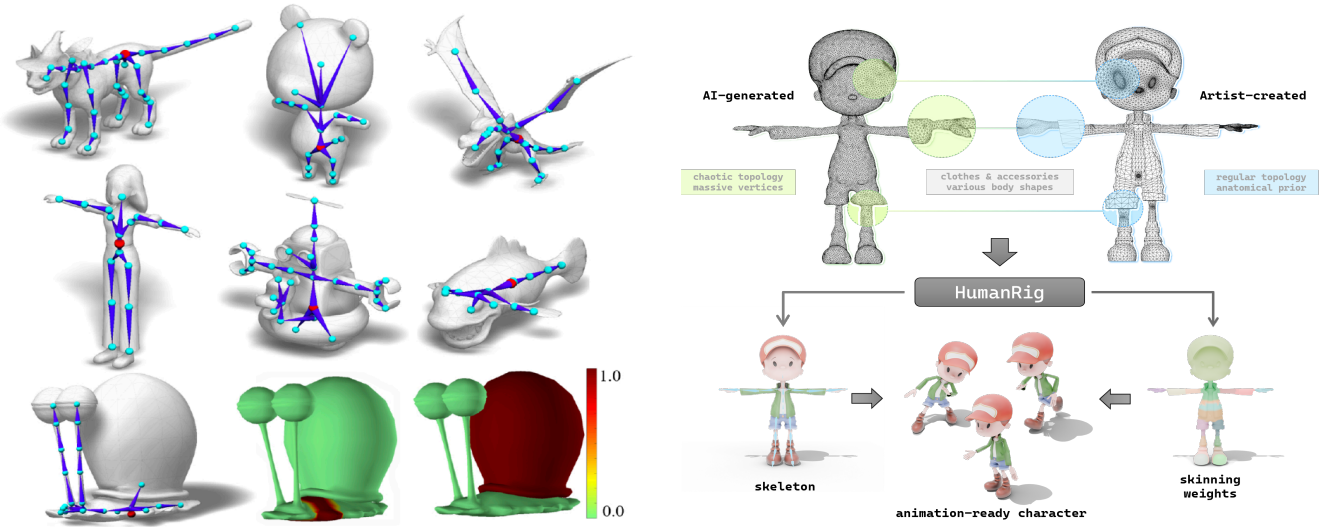


***Figure 1 Rigging comparison:*** *left shows artist-created rig with structured skeleton and meaningful skinning weights [5]; right shows automatic rigging framework that handles diverse mesh topology and outputs joints and skinning weights [2].*

## 1.2 Problem Statement

First, editing deformation fields in implicit neural representations typically requires retraining, making real-time interaction computationally expensive and impractical [12, 9]. Such methods also scale poorly with sequence length, as model complexity or embeddings often grow rapidly over time [10, 6].

Second, implicit deformation representations are difficult to convert into explicit skeletons or pose structures [11, 8], limiting their compatibility with standard animation pipelines such as Blender [13]. As a result, they struggle to support articulated motion editing and retargeting.

Third, temporal consistency remains challenging, as many neural rendering methods optimize frames independently, leading to jitter and inconsistent deformation in long sequences. In addition, skeleton-free representations often underrepresent high-frequency and non-rigid motions [12, 4].

Finally, there is a growing need for unified frameworks that bridge neural rendering and conventional animation pipelines, enabling editable and physically consistent 4D motion generation [2, 7]. In summary, the main challenges are:

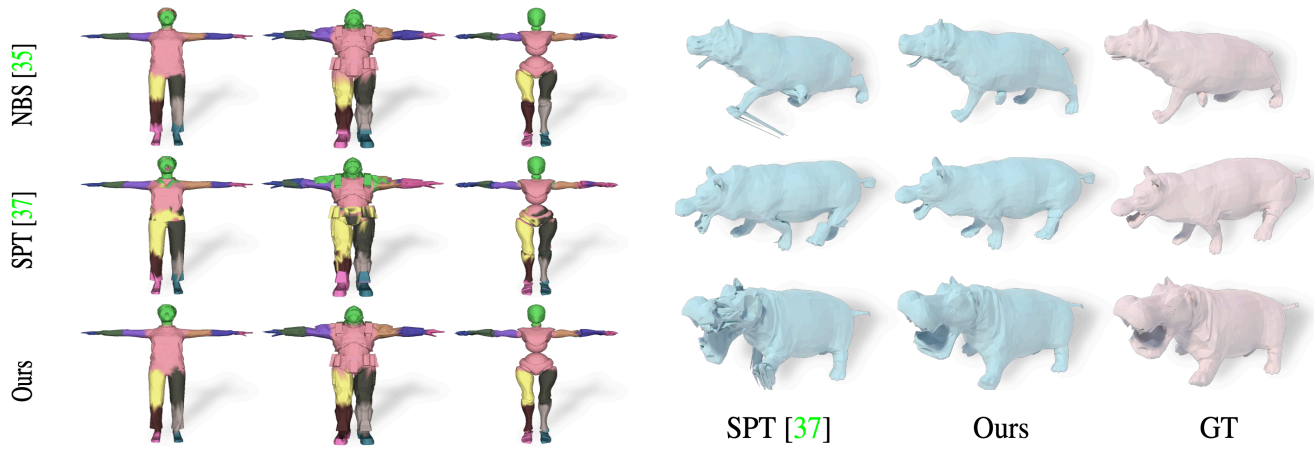


***Figure 2 Failure cases in implicit deformation and pose transfer:*** *Implicit representations are hard to map to skeletons or skinning weights [8], limiting controllable. Segmentation and pose transfer often produce articulation errors, especially in quadruped mouth and limb regions.*

- **Real-time editing limitation:** deformation fields require computationally expensive retrain [12].
- **Quadratic scaling:** parameter requirements grow rapidly with sequence length [10, 6].
- **Skeleton/pose conversion difficulty:** implicit representations do not map easily to standard skeletons or skinning weights [11, 8].
- **Temporal inconsistency:** per-frame optimization has jitter, low high-frequency fidelity [12, 4].
- **Pipeline integration challenge:** lack of the compatibility with Blender pipeline or standard animation workflows [13].

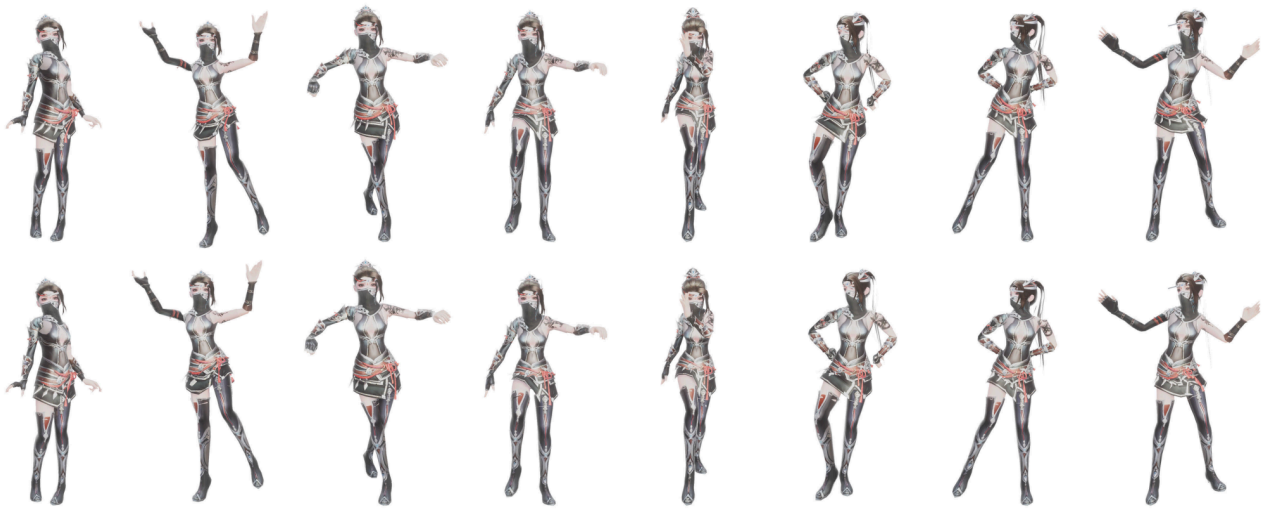

***Figure 3 Comparison of deformation results:*** *the top row shows ground-truth meshes present with artist-painted weights, while the bottom row shows results using our predicted weights[4].*

## 1.2 Motivation

Template-based methods work well on fixed or known topologies but do not generalize to unseen or irregular shapes. Template-free and mesh deformation methods are more flexible but often produce unstable skeletons and weak motion transfer, and are hard to integrate into standard animation pipelines. These issues are further amplified in large-scale generative settings with diverse 3D assets. In addition, real-world meshes vary significantly, where artist-created models are structured and joint-aware, while AI-generated meshes are often irregular and inconsistent. These limitations motivate a unified framework that can automatically extract robust skeletons and skinning weights from arbitrary 3D representations for reliable and editable motion generation.

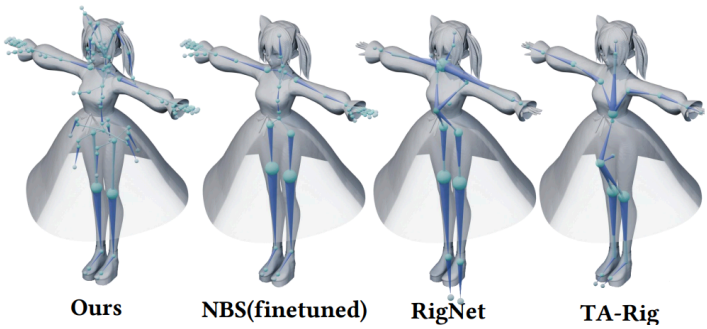


***Figure 4 Comparison of skeleton effects:*** *presents the difference between NBS (fine-tuned), RigNet, and TA-Rig on the VRoid dataset. Our UniRig generates more detailed and accurate skeletons [4].*

## 1.3 Problem Settings

In this work, we consider the problem of reconstructing and modeling dynamic 3D scenes under an extremely constrained setting. Specifically, we assume a **monocular setup** where only a single event camera is available, and the camera is **completely fixed without any displacement or rotation**.

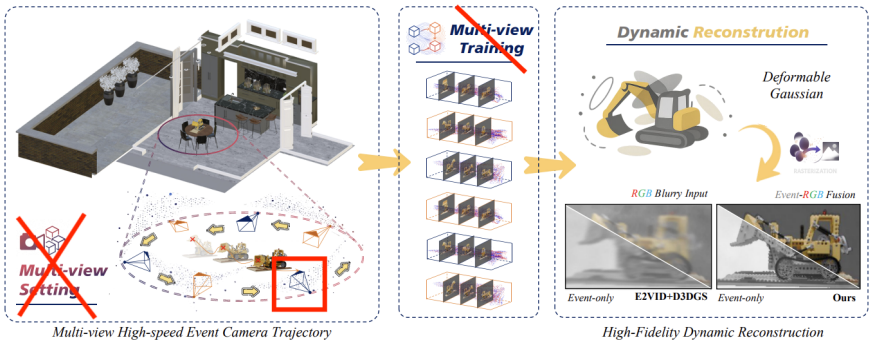


***Figure 5 Monocular Event Camera:*** *monocular viewpoint is reconstructed using 4DGS with event data.*

The input to our system is an **event-only stream**. Under this setting, our goal is to extend the formulation of **4D Gaussian Splatting (4DGS)** to a monocular, fixed-camera, event-driven scenario. (i) the absence of multi-view parallax leads to severe depth ambiguity, (ii) event streams provide only sparse and indirect observations of scene geometry, (iii) dynamic scene modeling must rely purely on temporal evolution without spatial baseline constraints.

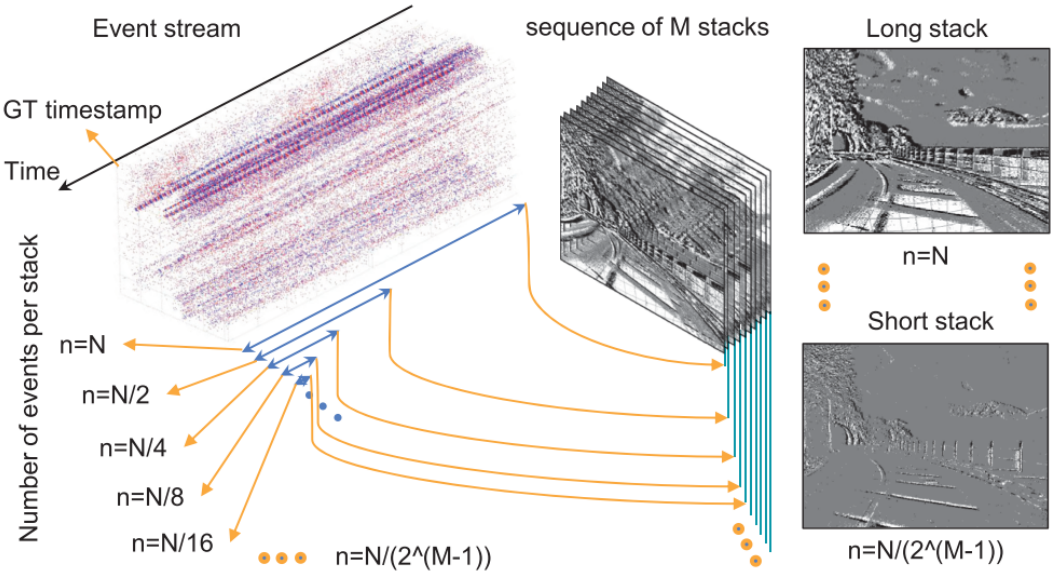


***Figure 6 Multi-Scale Event Stack Representation:*** *this demonstrates a method for dividing an event stream into multi-scale stacked sequences based on the number of events.*

# 2. Related Work

## 2.1 Skeleton-based Rigging Systems

Skeleton-based rigging provides structured motion priors for controllable deformation. **Classical Linear Blend Skinning (LBS)** enables efficient articulation. Recent automatic rigging systems such as RigNet, SMPL-based models, and UniRig aim to infer skeletal structures from unstructured geometry. But, these focus on mesh deformation and are not compatible with Gaussian representations.

***Figure 7 Effects of UniRig:*** *Examples from Rig-XL , demonstrating well-defined skeleton structures.*

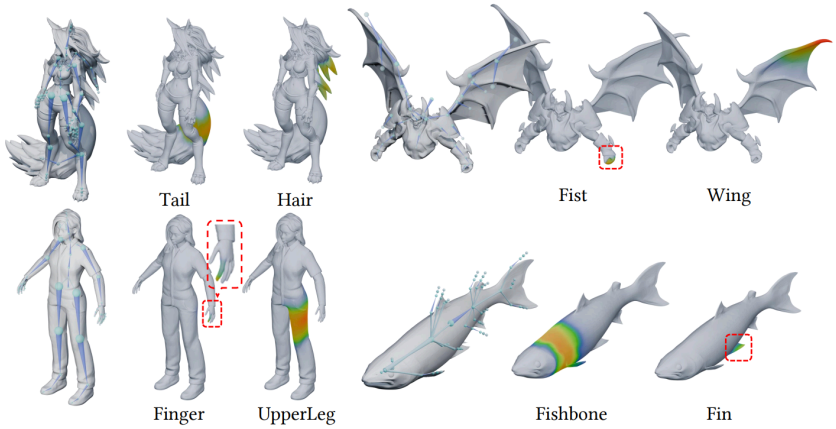


***Figure 8 Qualitative results of UniRig on various object categories:*** *The figure showcases the predicted skeletons, skinning weights (e.g., Tail, Hair, Fist, Wing, Finger, UpperLeg, Fishbone, Fin).*

The method still exhibits several limitations. First, fine-grained regions such as fingers, fists, and fins may suffer from over-smoothed skinning weights, leading to blurry boundaries and stretching artifacts during animation. Second, under complex topologies, the predicted skeleton may contain redundant or misaligned joints, increasing deformation instability and the risk of mesh self-intersection. Finally, for

non-humanoid appendages such as hair and wings, the model struggles with extreme poses, often producing less smooth mesh transitions under large deformations.

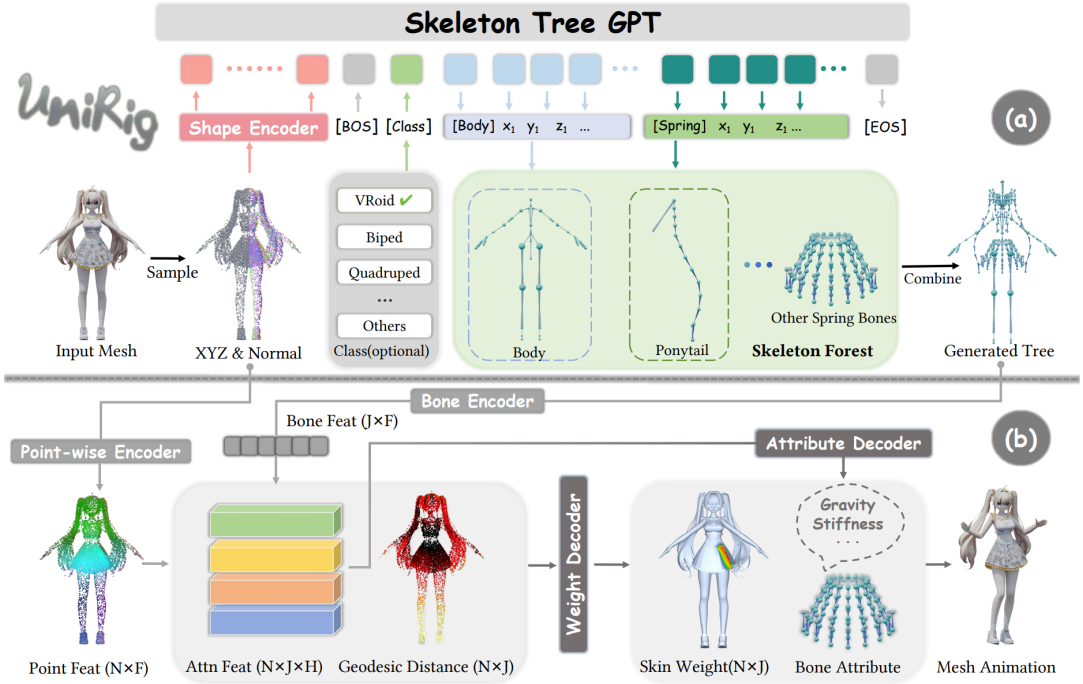


***Figure 9: Overview of UniRig Framework*:** *The framework consists of two main stages:* ***(a) Skeleton Tree Prediction****, which processes mesh-sampled point cloud through a Shape Encoder and a Skeleton Tree GPT to decode a hierarchical skeleton structure; and* ***(b) Skin Weight Prediction****, which fuses point-cloud and bone features via a Bone-Point Cross Attention mechanism.*

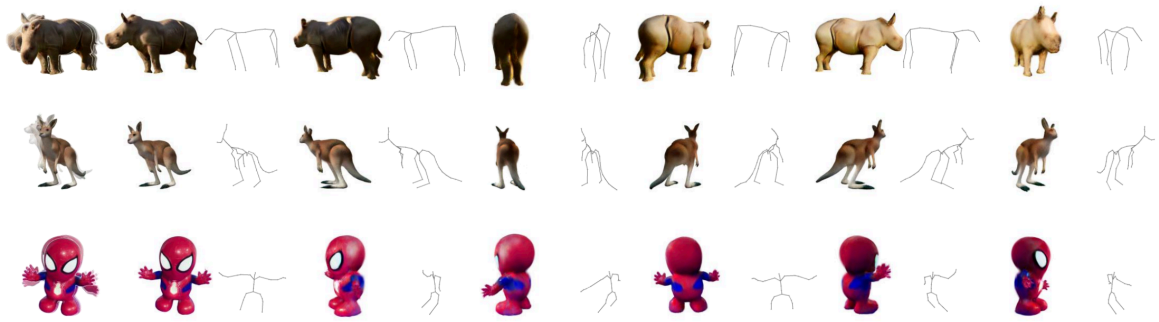

***Figure 10: Effect of Visualizing 4D Object Motion with Skeleton Poses****: it presents generated 4D motion and corresponding skeleton poses, with viewpoint and time progressing from left to right.*

**Table 1**

*Comparison of UniRig with prior work in automatic rigging.*

| Method | Template Based | Template Free | Automation | Multi Cat. | Cost Time |
|---|---|---|---|---|---|
| RigNet [5] | × | ✓ | Automated | ✓ | 1s–20min[†] |
| NBS [3] | ✓ | × | Automated | × | 1 s |
| TaRig [14] | ✓ | ✓ | Automated | × | 30 s |
| Anything World [15] | ✓ | ✓ | Semi-Automated | ✓ | 5 min |
| Tripo [16] | ✓ | ✓ | Automated | ✓[*] | 2 min |
| Meshy [17] | ✓ | × | Semi-Automated | × | 1–2 min |
| Accurig [18] | ✓ | × | Semi-Automated | × | 1 min |
| UniRig (Ours) | ✓ | ✓ | Automated | ✓ | 1–5 s |

[*] Tripo supports only human and quadruped categories.

[†] Inference time depends on number of bones and model complexity.

## 2.2 Monocular Videos with 4D Reconstruction

Monocular 4D reconstruction aims to recover dynamic geometry and motion from single-view videos. Existing methods often rely on neural deformation fields or per-frame optimization.

Despite progress, these methods struggle with temporal consistency, structural rigidity, and long-term deformation stability, especially under complex articulated motion.

## 2.3 Key Limitations of Existing Methods

Despite recent progress in 3D Gaussian Splatting (3DGS) and its dynamic extensions (e.g., 4DGS), existing methods still face fundamental limitations when applied to monocular, event-based, and fixed-camera scenarios. We summarize the key challenges as follows:

### 2.3.1 Multi-view Dependency and 4DGS Assumption Breakdown

Most existing 4D Gaussian Splatting frameworks are built upon the assumption of **multi-view inputs** with sufficient spatial parallax. However, in a monocular fixed-camera setting, this assumption no longer holds. The lack of translational or rotational camera motion leads to a severe collapse of geometric constraints, making traditional multi-view optimization inapplicable.

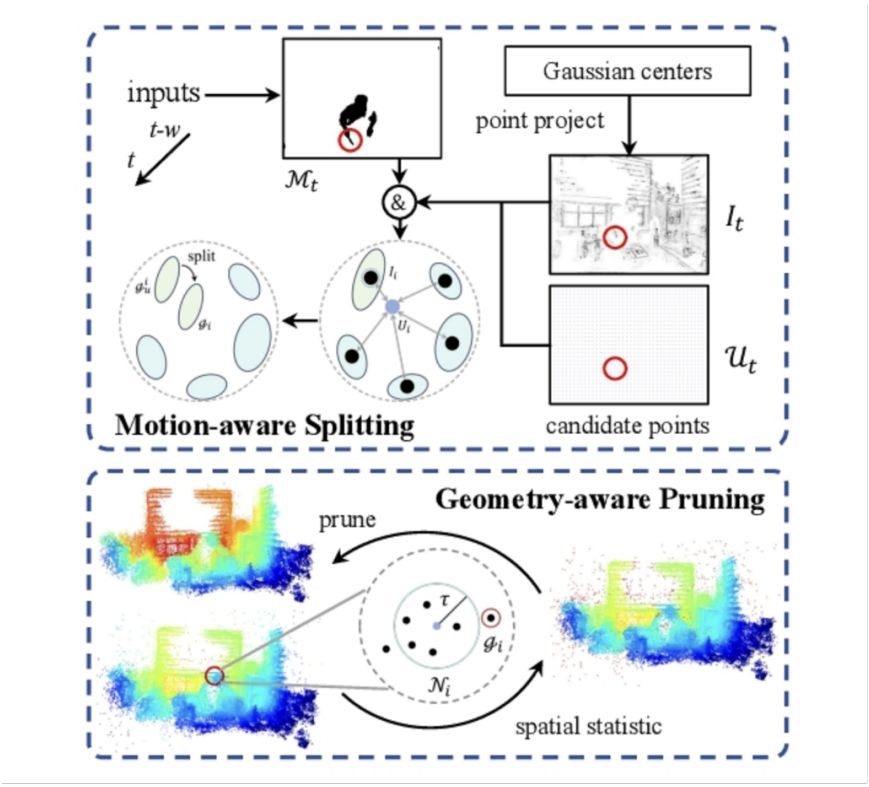


***Figure 11 Motion-aware Splitting and Geometry-aware Pruning Framework:*** *this figure illustrates the motion-aware splitting and geometry-aware pruning process based on Gaussian central projection and candidate point constraints, which is used to improve the structural rationality of dynamic scene.*

### 2.3.2 Severe Depth Ambiguity under Fixed Camera

Without parallax cues, purely temporal observations are insufficient to recover consistent 3D geometry. It leads to geometric degradation, unstable depth estimation, and reduced reconstruction.

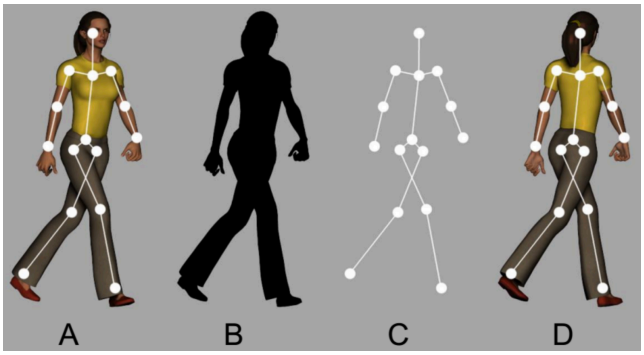


***Figure 12 Strong depth prediction ambiguity:*** *Under a stationary monocular setup, the absence of spatial baseline results in strong depth ambiguity*

### 2.3.3 Non-Separation of Static and Dynamic Gaussians

Existing dynamic Gaussian methods typically model the entire scene using a unified deformable field, without explicitly distinguishing static and dynamic components. Since real-world scenes are often dominated by static regions (approximately 90%), this design introduces unnecessary deformation computation, increases noise accumulation, and reduces optimization efficiency.

***Figure 13 Dynamic-Aware GS Training and Static Scene Reconstruction from Novel View:*** *this figure illustrates the overall framework for improving static scene reconstruction from a new perspective by introducing static-controlled alpha-blending for dynamic-aware Gaussian training.*

### 2.3.4 Limitations of Event-Only Observations

Event-based sensors provide only asynchronous brightness changes rather than dense intensity measurements. While they offer high temporal resolution, they inherently lack absolute appearance information and spatial gradients. In a fixed-camera configuration, this limitation becomes more severe, as events alone are insufficient to recover stable geometry without additional structural priors.

# 3. Proposed Method

## 3.1 Overview

We propose a hybrid framework that combines: (i) canonical 3D Gaussian representation, (ii) skeleton-driven rigid deformation, and (iii) neural residual non-rigid deformation.

Given a monocular video, we first construct a canonical Gaussian field, then apply skeleton-based motion via forward kinematics and LBS, followed by a neural residual deformation field to capture non-rigid details. Finally, the deformed Gaussians are rendered in a differentiable pipeline optimized via

reconstruction loss and score distillation.

### 3.1.1 Canonical Space Gaussian Representation

We define a canonical Gaussian field:

$$\mathcal{G}_c = \{\mathbf{p}_c^i, \mathbf{q}_c^i, \mathbf{s}^i, \sigma^i, \mathbf{c}^i\}_{i=1}^N. \tag{1}$$

Each Gaussian represents a deformable primitive encoding geometry and appearance, where the center $\mathbf{p}_c^i$, rotation $\mathbf{q}_c^i$, scale $\mathbf{s}^i$, opacity $\sigma^i$, and spherical harmonic coefficients $\mathbf{c}^i$ jointly describe its spatial and radiometric properties.

The covariance matrix is formulated as:

$$\Sigma^i = \mathbf{R}(\mathbf{q}_c^i)\text{diag}(\mathbf{s}^i)\text{diag}(\mathbf{s}^i)^T\mathbf{R}(\mathbf{q}_c^i)^T. \tag{2}$$

it allows each Gaussian to model anisotropic deformation aligned with orientation in 3D space.

### 3.1.2 Skeleton Definition and Forward Kinematics

We define the skeleton as: $\mathcal{S} = (\mathcal{J}, \mathcal{E})$ , where $\mathcal{J} = \{J_k\}_{k=1}^B$ denotes joints and $\mathcal{E}$ defines their connectivity structure. Each joint is associated with a rotation parameter, which together define the full articulated pose at time $t$:

$$\theta_t^k \in SO(3), \quad \Theta_t = \{\theta_t^k\}_{k=1}^B, \tag{3}$$

The global transformation of each joint is computed via forward kinematics:

$$\mathbf{B}_k(\Theta_t) = \prod_{j \in \mathcal{A}(k)} \mathbf{T}(\theta_t^j), \tag{4}$$

where $\mathcal{A}(k)$ denotes the set of ancestor joints in the kinematic tree. This hierarchical formulation ensures that local joint rotations are properly accumulated into global transformations.

Each transformation matrix is defined as:

$$\mathbf{T}(\theta_t^j) = \begin{bmatrix} \mathbf{R}(\theta_t^j) & \mathbf{t}_j \\ 0 & 1 \end{bmatrix}. \tag{5}$$

### 3.1.3 KNN Skinning and LBS Deformation

To associate each Gaussian with the skeleton, we compute inverse-distance based skinning weights:

$$w_{ik} = \frac{(d_{ik} + \epsilon)^{-1}}{\sum_{k'}(d_{ik'} + \epsilon)^{-1}}, \quad d_{ik} = \|\mathbf{p}_c^i - J_k\|_2. \tag{6}$$

This formulation assigns higher influence to spatially closer joints, ensuring smooth spatial blending of skeletal influence. Using these weights, by **Linear Blend Skinning (LBS)** to deform Gaussian centers:

$$\mathbf{p}_r^i = \sum_{k=1}^{B} w_{ik} \mathbf{B}_k(\Theta_t) \mathbf{p}_c^i. \tag{7}$$

This step produces a rigidly articulated deformation that preserves skeletal structure while maintaining continuity across Gaussians. The resulting rigidly deformed Gaussian field is:

$$\mathcal{G}_r = \mathcal{F}_{\text{LBS}}(\mathcal{G}_c, \Theta_t). \tag{8}$$

## 3.2 Joint 2D–3D Closed-Loop Optimization

### 3.2.1 Non-Rigid Residual Field (HexPlane)

To model fine-grained non-rigid deform beyond skeletal motion, it has residual deformation field:

$$\Delta\mathcal{G} = \mathcal{F}_\phi(x, y, z, t). \tag{9}$$

This field captures residual geometric and appearance variations that cannot be explained by skeletal articulation alone. We adopt a HexPlane factorization to efficiently represent 4D spatiotemporal features:

$$f(x, y, z, t) = \bigoplus_{P \in \{XY, XZ, YZ, XT, YT, ZT\}} \text{Interp}(P, \pi_P(x, y, z, t)). \tag{10}$$

This decomposition projects 4D coordinates into multiple 2D planes, improving both efficiency and generalization in modeling dynamic scenes. **Residual deformation** is predicted, where outputs corrections to position, rotation, and scale of each Gaussian:

$$(\Delta\mathbf{p}, \Delta\mathbf{q}, \Delta\mathbf{s}) = \text{MLP}(f), \tag{11}$$

The final Gaussian field is obtained by combining rigid and non-rigid deformation:

$$\mathcal{G}_o = \mathcal{G}_r + \Delta\mathcal{G}. \tag{12}$$

### 3.2.2 Differentiable Rendering

The deformed Gaussian field is rendered into an image via a differentiable renderer:

$$I = \mathcal{R}(\mathcal{G}_o, \pi), \tag{13}$$

where $\pi$ denotes the camera pose.

This rendering process is fully differentiable, allowing gradients to propagate back to both geometric and skeletal parameters:

$$\frac{\partial I}{\partial \theta} = \frac{\partial I}{\partial \mathcal{G}} \frac{\partial \mathcal{G}}{\partial \Theta} \frac{\partial \Theta}{\partial \theta}. \tag{14}$$

### 3.2.3 Score Distillation Sampling (SDS)

We adopt a diffusion-based supervision signal using Score Distillation Sampling (SDS). The rendered image is perturbed with Gaussian noise:

$$z_t = \alpha_t I + \sigma_t \epsilon, \tag{15}$$

where $\epsilon$ is standard Gaussian noise.

The SDS objective aligns rendered results with a pretrained diffusion prior:

$$\nabla_\Theta \mathcal{L}_{SDS} = \mathbb{E}_{t,\epsilon} \left[ w(t)(\hat{\epsilon}_\phi(z_t) - \epsilon) \frac{\partial I}{\partial \Theta} \right]. \tag{16}$$

The final optimization objective combines reconstruction, diffusion guidance, and regularization:

$$\mathcal{L} = \mathcal{L}_{SDS} + \lambda_{rgb} \| I - I_{gt} \|_2^2 + \lambda_{tv} \| \nabla f \|_1. \tag{17}$$

### 3.2.4 Learnable Skinning Stability

We replace fixed inverse-distance weights with a learnable skinning function:

$$w_{ik} = \text{softmax}(g_\psi(\mathbf{p}_c^i, J_k)). \tag{18}$$

This allows the model to learn adaptive joint influence beyond geometric proximity. The gradient with respect to $\psi$ is:

$$\frac{\partial \mathcal{L}}{\partial \psi} = \sum_{i,k} \frac{\partial \mathcal{L}}{\partial \mathbf{p}_r^i} \cdot \mathbf{B}_k(\Theta_t) \mathbf{p}_c^i \cdot \frac{\partial w_{ik}}{\partial \psi}. \tag{19}$$

However, this formulation introduces optimization instability. When a Gaussian is extremely close to a joint, the gradient may explode:

$$d_{ik} \to 0 \Rightarrow \frac{\partial w_{ik}}{\partial \psi} \to \infty, \tag{20}$$

while in cases of dominant joints, gradients may vanish:

$$w_{ik} \approx \mathbb{1}(k = k^*) \Rightarrow \nabla_\psi \to 0. \tag{21}$$

To mitigate temporal inconsistency, we introduce a smoothness constraint:

$$\mathcal{L}_{temp} = \sum_t \| W^t - W^{t+1} \|_F^2. \tag{22}$$

## 3.3 Core Innovation Overview

To address the limitations of existing 4D Gaussian Splatting methods under monocular, fixed-camera, and event-only settings, we propose a unified framework consisting of four key innovations. These components jointly enable stable geometry reconstruction, temporal modeling, and event-consistent supervision.

### 3.3.1 Static Background Decoupling and Temporal Optical Flow-Driven Representation

We first introduce a dual-branch Gaussian representation that explicitly decouples the scene into **static background Gaussians** and **dynamic foreground Gaussians**.

For static regions, Gaussians are permanently anchored in space, and the deformation network is disabled. Instead, they are optimized through long-term event accumulation to ensure geometric consistency and prevent spatial drift.

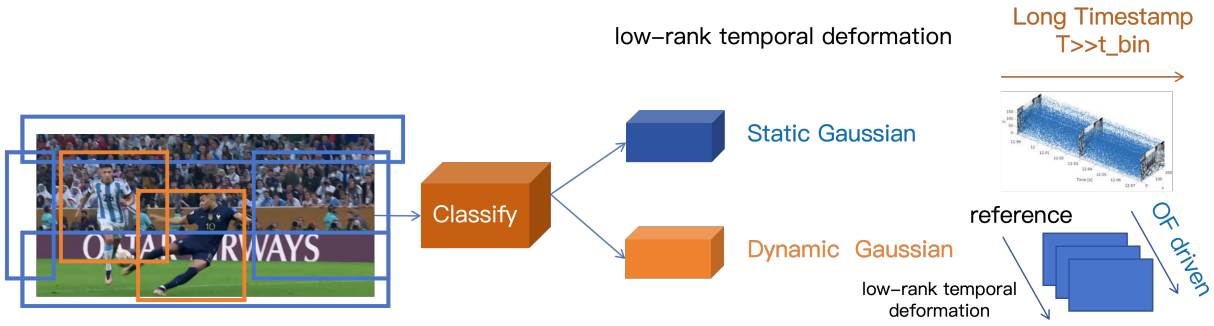


***Figure 14 Static Background Decoupling and Temporal Optical Flow-Driven Dual-Branch:*** *(1) static Gaussian disables the deformation network, using only long-term event accumulation for geometric refinement, permanently fixing the spatial position; (2) dynamic Gaussian is driven by event data asynchronous optical flow to get a low-rank temporal deformation basis.*

For dynamic regions, we model temporal evolution using an **event-asynchronous optical flow prior**, which induces a low-rank temporal deformation basis. This design eliminates the need for frame-by-frame pose optimization and significantly reduces redundancy in temporal modeling.

Formally, the representation can be expressed as:

$$\mathcal{G} = \mathcal{G}_{static} \cup \mathcal{G}_{dynamic}, \quad \mathcal{G}_{static} \cap \mathcal{G}_{dynamic} = \emptyset \tag{23}$$

where static Gaussians remain invariant over time, while dynamic Gaussians evolve according to optical-flow-guided deformation fields.

### 3.3.2 Fixed-Pose Temporal Stereo Parallax Modeling

To overcome the absence of multi-view supervision in monocular settings, we propose a novel **temporal stereo formulation**. Instead of relying on spatial baseline, we treat different timestamps as virtual viewpoints $t_1, t_2, \ldots, t_n$:

$$t_1, t_2, \ldots, t_n \quad \rightarrow \quad \text{pseudo multi-view observations} \tag{24}$$

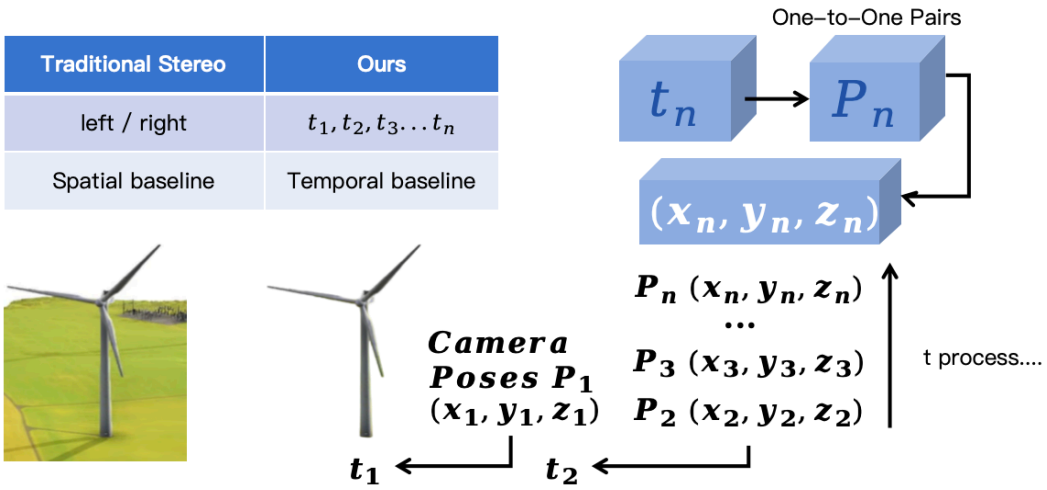


***Figure 15 Temporal Stereo Matching via One-to-One Time–Pose Pairs:*** *this demonstrates a method for temporal stereo modeling by replacing the spatial baseline in traditional binocular vision with a temporal baseline and achieving one-to-one correspondence between camera poses at different times.*

Each time step corresponds to a one-to-one geometric correspondence:

$$P_t = \Pi(\mathcal{G}, t), \quad (x_t, y_t, z_t) \tag{25}$$

This formulation converts temporal variation into a substitute for spatial parallax, enabling 4DGS optimization under a fixed camera setting. In this way, motion provides implicit structure cues that replace classical stereo geometry.

### 3.3.3 Gaussian Space Adaptive Event Thresholding with Co-Optimization

We introduce a **learnable, region-aware event threshold mechanism** that enables adaptive separation of static and dynamic Gaussians under event-based supervision.

Unlike conventional methods that employ a global and fixed event-triggering threshold, which treats all spatial regions uniformly, our approach assigns a **Gaussian-specific sensitivity parameter** $C_i$ to each 3D Gaussian primitive. This allows model to encode spatial heterogeneity in motion sensitivity.

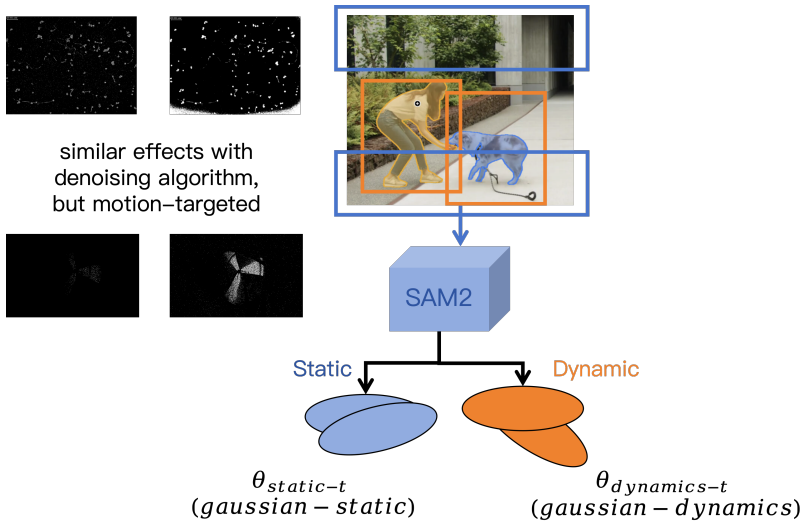


***Figure 16 Region-Aware Event Threshold for Static–Dynamic Gaussian Separation:*** *it demonstrates that by introducing a learnable region-aware event threshold and assigning adaptive motion sensitivity to each Gaussian, static Gaussians and dynamic Gaussians can be separated more accurately.*

To further improve structural consistency, we introduce a **region-aware initialization strategy** using segmentation priors (e.g., SAM2). The scene is decomposed into three semantic-consistent regions:

- Static background regions: assigned high thresholds to suppress noise-induced activations;
- Dynamic object regions: assigned low thresholds to enhance motion sensitivity;
- Boundary regions: assigned intermediate thresholds to stabilize transition zones.

Unlike semantic segmentation, our formulation does not rely on object categories. Instead, all objects are unified under a motion-centric representation, where the threshold reflects **temporal variability rather than semantic identity**.

This design allows the model to dynamically adapt event sensitivity across different spatial regions, effectively suppressing static drift while enhancing motion-aware responsiveness in dynamic areas. As a result, the system achieves a more robust decomposition of static and dynamic components in event-driven 4D Gaussian representations.

We further introduce a spatially adaptive event threshold mechanism, where each Gaussian is assigned a learnable sensitivity parameter $C_i$:

$$E(x,t) = \mathbb{1}\left(|\Delta \log I(x,t)| > C_i\right) \tag{26}$$

Different scene regions are assigned distinct thresholds:

- Static background: high threshold (insensitive to noise)
- Dynamic objects: low threshold (high sensitivity to motion)
- Boundary regions: medium threshold

To improve adaptability, we jointly optimize Gaussian parameters and thresholds:

$$\mathcal{L} = \mathcal{L}_{render} + \mathcal{L}_{event}(C_i) + \mathcal{L}_{reg}(C_i) \tag{27}$$

This formulation allows the system to dynamically adjust event sensitivity according to spatial structure complexity, leading to improved robustness under heterogeneous motion patterns.

#### 3.3.4 Event-Based Temporal Derivative Supervision and Three-Stage Training Strategy

Finally, we exploit the intrinsic relationship between event generation and temporal intensity changes by supervising the time derivative of Gaussian rendering outputs:

$$E(x,t) \propto \frac{\partial I(x,t)}{\partial t} \tag{28}$$

This enables continuous-time supervision without requiring explicit frame-based reconstruction or auxiliary networks such as E2VID.

We further design a three-stage training pipeline, this staged optimization strategy ensures stable convergence while preserving fine-grained dynamic motion modeling capability:

1. **Static Background Pre-Reconstruction:** Accumulate early event streams and freeze background Gaussians to establish stable geometry.
2. **Dynamic Gaussian Incremental Modeling:** Introduce dynamic Gaussians only in motion-active regions guided by event signals.
3. **Global Joint Optimization:** Fix static structure and jointly refine dynamic deformation fields for temporal consistency.

## 4. Experimental Setup

We present a comprehensive experimental setup to evaluate two representative systems: **UniRig** and **SkeletonGaussian**. These systems cover different aspects of geometric learning and dynamic scene modeling, including skeleton-based rigging, Gaussian-based representations, and monocular event/video-driven supervision.

## 4.1 P-MASK Segmentation Algorithm

To address the limitations of existing segmentation frameworks under event-driven and dynamic scenarios, we propose **P-MASK**, a language-guided and memory-consistent segmentation model designed to improve object tracking stability and reduce attention drift.

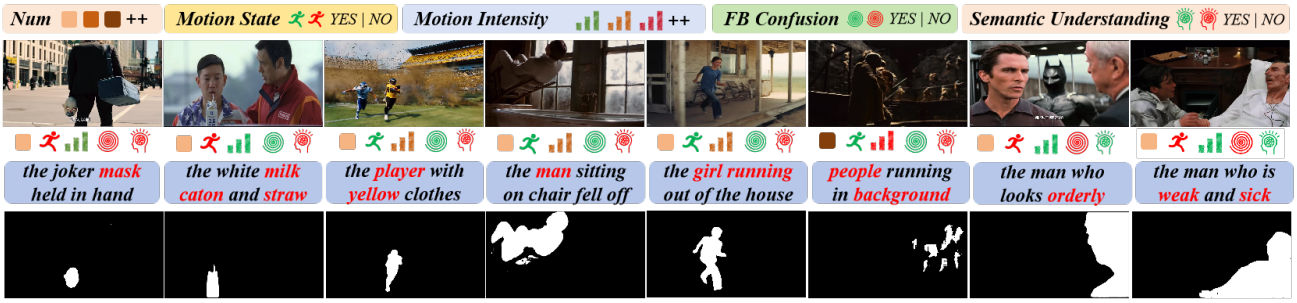


***Figure 17 Multi-Dimensional Event Understanding and Motion-Aware Semantic P–MASK:*** *it presents an event visual understanding framework that combines multiple dimensions such as target quantity, motion state, motion intensity, fore/background obfuscation, semantic understanding.*

### 4.1.1 P-MASK: Language-Guided Segmentation with Memory Correction

To overcome these issues, we propose **P-MASK**, which introduces two key improvements:

- **Language-guided object targeting:** Unlike SAM2, P-MASK supports natural language queries, enabling flexible and semantic-aware object selection without manual mask initialization.
- **Memory-consistent correction mechanism:** We introduce a memory refinement strategy that mitigates accumulated tracking errors over time, improving robustness in long video sequences.

Formally, given an input frame $I_t$ and language prompt $L$, P-MASK predicts a mask $M_t$ as:

$$M_t = \mathcal{F}_{pmask}(I_t, L, H_{t-1}), \tag{29}$$

where $H_{t-1}$ denotes historical memory embeddings used for temporal consistency correction.

### 4.1.2 Robustness in Complex Dynamic Scenes

P-MASK demonstrates strong robustness in challenging scenarios. We consider several challenging scenarios, including fast-moving objects that exhibit motion blur or abrupt displacement.

## 4.2 UniRig: Skeleton Extraction and Structural Limitations

In this section, we evaluate the performance of UniRig on skeleton extraction tasks and analyze its applicability boundaries under complex real-world geometries.

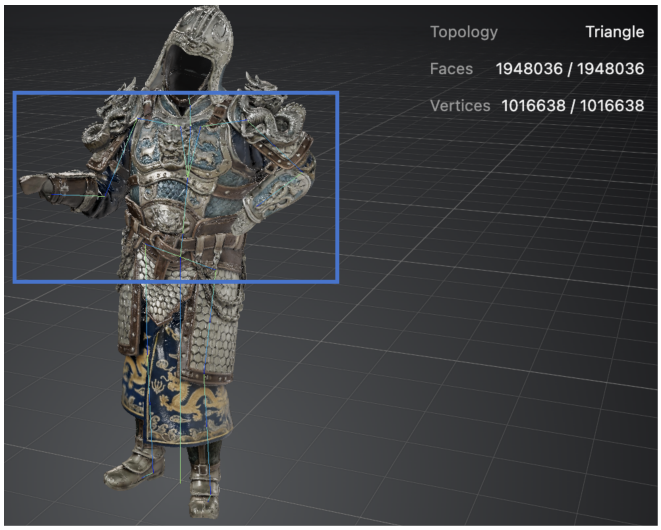


***Figure 18 Skelton Extraction Visualization:*** *this is one model with Unirig effects, skelton visualzation may be small, but it still shows the effective demostration of skelton extraction*

### 4.2.1 Skeleton Extraction on Structured and Unstructured Objects

UniRig demonstrates strong capability in extracting skeleton representations from articulated and semi-articulated objects, particularly in structured 3D assets such as humanoid armor models.

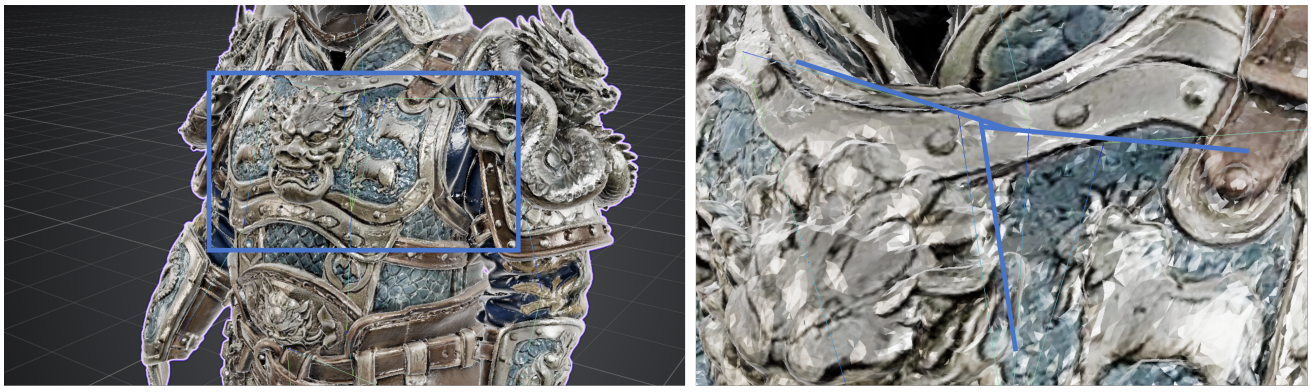

***Figure 19 Skelton Extraction Visualization:*** *this is one model with Unirig effects, skelton visualzation may be small, but it still shows the effective demostration of skelton extraction*

For humanoid-like objects, UniRig can recover a plausible skeletal topology even under complex surface geometry. However, the quality of skeleton visualization may vary depending on mesh density and surface noise, while still preserving consistent structural decomposition.

### 4.2.2 Wind Turbine Case: Dynamic vs Static Decomposition

We further evaluate UniRig on wind turbine scenes, which introduce a challenging scenario involving thin structures and rigid motion. The results show that UniRig can effectively separate the scene into:

- **Dynamic components:** rotating turbine blades driven by motion cues;
- **Static components:** supporting tower structure with minimal deformation.

This decomposition demonstrates that motion-aware segmentation can be achieved even without explicit skeletal priors, relying purely on geometric and temporal cues.

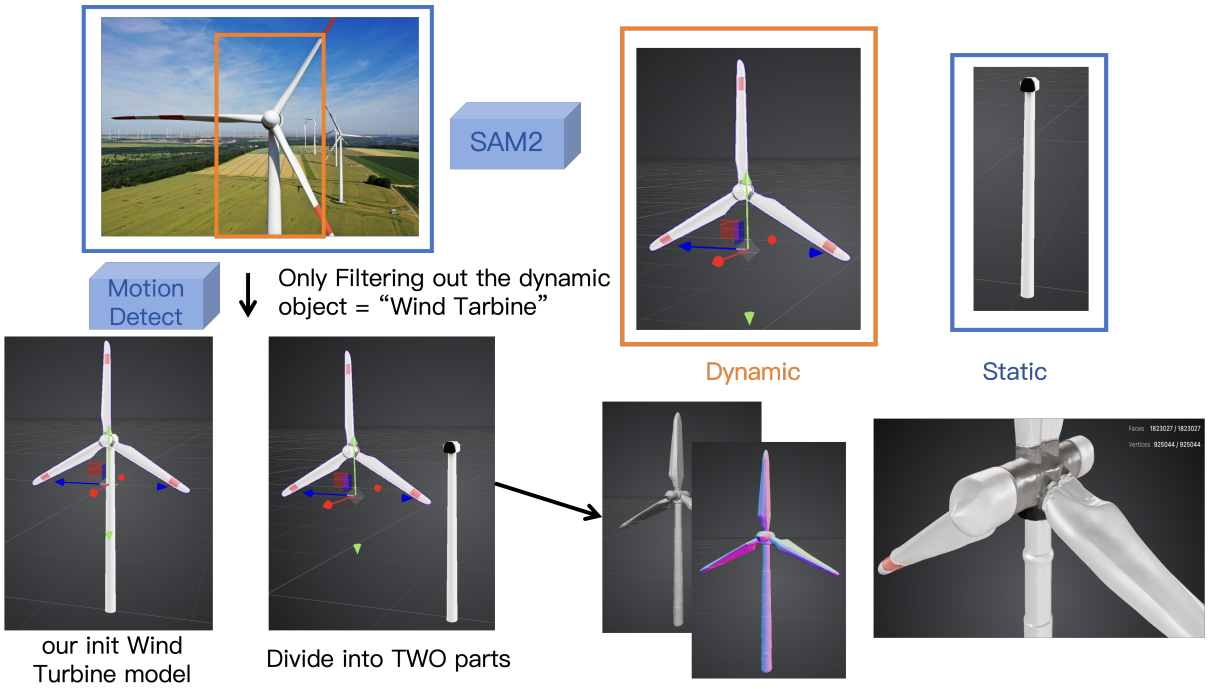


***Figure 20 Wind Turbine Detection Visualization:*** *the wind turbine is divided into dynamic and static state gaussian and our focus lays on blade.*

### 4.2.3 Failure Case: Skeleton Extraction Limitations

Despite its success in structured objects, UniRig exhibits clear limits in some challenging cases:

We consider several challenging scenarios, including non-chain structures such as wind turbine blades and thin sheets, which are often incorrectly modeled as single rigid parts rather than articulated

chains; highly rigid industrial objects where no meaningful skeletal hierarchy exists, making skeleton extraction unnecessary or ill-posed; non-smooth surfaces where reconstruction artifacts and geometric noise lead to unstable joint placement and inaccurate topology inference; and highly branching structures such as trees or bushes, which introduce significant ambiguity in skeleton graph construction.

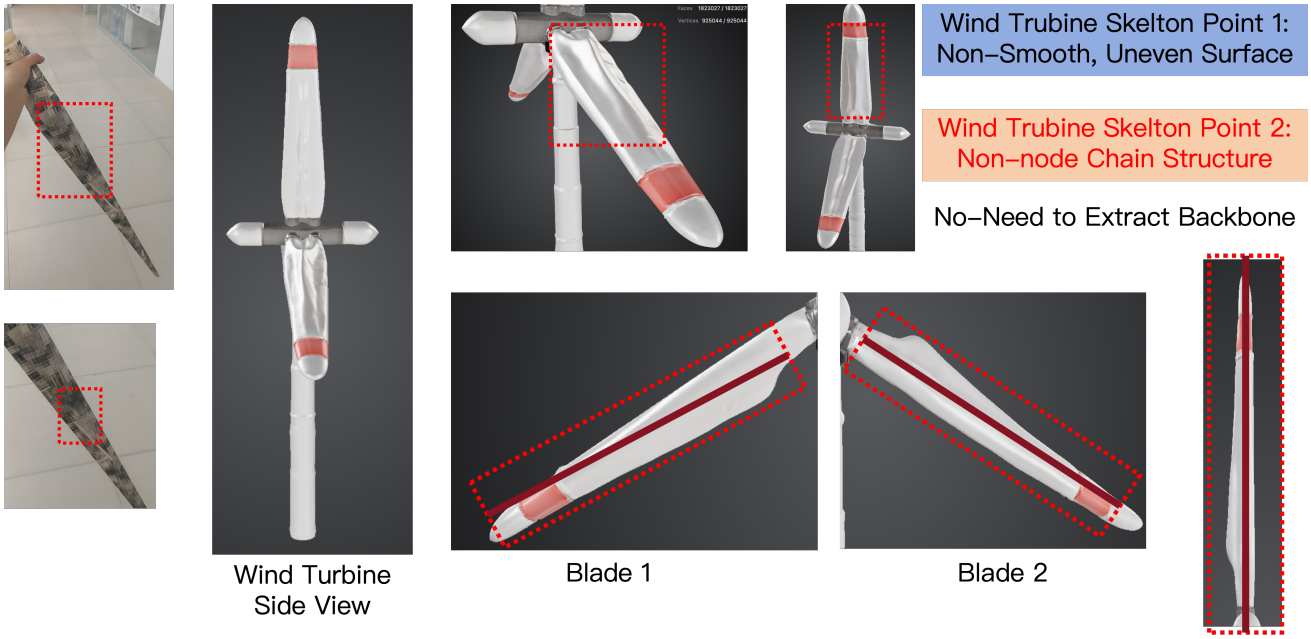


***Figure 21 Wind Turbine Skeleton Reconstruction Failure Analysis:*** *this demonstrates the difficulties in anomaly detection and frame extraction during wind turbine frame reconstruction caused by surface irregularities, wrinkled structures, and non-node chain-like features.*

Wind turbine blades are particularly problematic: due to their thin-sheet structure, the model collapses multiple blades into a single central node with three branching directions, failing to capture true mechanical structure.

### 4.2.4 Wind Turbine Analysis: Why Skeletonization Fails

We further analyze the failure mechanism in wind turbine scenarios. Unlike humanoid objects designed for T-pose or A-pose representations, wind turbine blades do not follow a kinematic chain structure. Instead, they form a rigid planar geometry attached to a central hub. This leads to two key issues:

- **Incorrect structural prior:** skeleton-based models assume articulated joints, which do not exist in blade structures.
- **Degenerated topology:** the model collapses the structure into a central hub with pseudo-branches, misrepresenting physical connectivity.

Moreover, coupled motion analysis reveals that the true rotational dynamics are centered at the hub rather than distributed along blade geometry. Therefore, skeleton-based representation is fundamentally unsuitable for such rigid rotational systems.

**Table 2**

*Experimental setup of UniRig framework.*

| Setting | VRoid | Mixamo | Rig-XL | Notes |
|---|---|---|---|---|
| Dataset Size | 2,061 | - | 14,611 | Dataset composition |
| Sampling Ratio | 25% | 5% | 70% | Long-tail balancing |
| Augmentation | Rotation / Scaling / Motion perturbation | | | $p = 0.4$–$0.6$ |
| Stage-1 Model | OPT-125M + 3DShape2VecSet | | | Skeleton prediction |
| Stage-1 Training | 8×A100, 3 days, AdamW | | | LR: $1e^{-3} \rightarrow 2e^{-4}$ |
| Stage-2 Model | Frozen Point Transformer V3 | | | Skinning prediction |
| Stage-2 Training | 8×A100, 1 day | | | 16K points per mesh |
| Baselines | RigNet, NBS, TA-Rig, Meshy, Accurig, Tripo | | | Comparison methods |
| Metrics | J2J / J2B / B2B / L1 / L2 | | | Geometry accuracy |

## 4.3 SkeletonGaussian

**Table 3**

*Experimental setup of SkeletonGaussian framework.*

| Setting | Stage-1 | Stage-2 | Stage-3 | Notes |
|---|---|---|---|---|
| Dataset | Consistent4D + Objaverse-20 | | | 24 videos, ∼32 frames each |
| Initialization | Zero123++ views | 10K Gaussians | - | Canonical space init |
| Optimization Steps | 1500 | 2500 | 7000 | Multi-stage pipeline |
| Rigid Modeling | - | LBS + sliding window=3 | - | $5e^{-5} \rightarrow 5e^{-6}$ |
| Non-rigid Model | - | - | HexPlane + MLP | Temporal deformation |
| Loss Weights | $L_{MV-SDS} = 1$, $L_{rec} = 2 \times 10^4$, $L_{mask} = 10^3$ | | | Joint optimization |
| Baselines | Consistent4D, STAG4D, 4DGen, DreamGaussian4D | | | SOTA comparison |
| Metrics | CLIP / LPIPS / FVD | | | Image + video quality |
| Hardware | RTX 3090 | | | 1h training, 150 FPS |

## 4.4 Dense Optical Flow from Event

In this experiment, we investigate the problem of estimating dense optical flow directly from event streams. Unlike traditional frame-based methods, event cameras provide asynchronous spatio-temporal signals that encode intensity changes, making optical flow estimation fundamentally different and more challenging. Given an input event stream $E(x, t)$, the goal is to predict a dense optical flow field:

$$\mathbf{F}(x) = (u(x), v(x)), \tag{30}$$

where $(u, v)$ represent pixel-wise motion vectors derived solely from event-based observations without relying on RGB frames.

### 4.4.1 Event-to-Flow Learning Formulation

We formulate optical flow prediction as a mapping:

$$\mathcal{F}_\theta : E(x, t) \rightarrow \mathbf{F}(x), \tag{31}$$

where the network learns motion priors directly from asynchronous event signals.

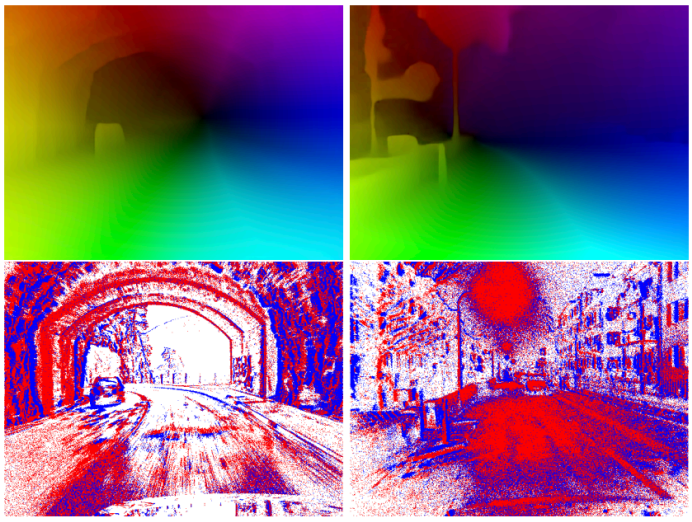

***Figure 22 Dense Optical Flow Estimation from Event-based Input:*** *the visuals demonstrate how dense optical flow fields can be recovered from event data input.*

### 4.4.2 Integration with Gaussian Splatting Pipeline

The predicted dense optical flow is further integrated into our Gaussian representation framework. Specifically, it provides a motion prior for dynamic Gaussian deformation, enabling low-rank temporal modeling without requiring frame-based supervision.

## 4.5 SC-GS: Sparse-Controlled GS

In this experiment, we propose a Sparse-Controlled Gaussian Splatting (SC-GS) framework that introduces a compact set of sparse control points to explicitly drive the motion of surrounding 3D Gaussians. Instead of directly optimizing dense Gaussian deformation fields, we first extract a small number of representative control points using a **KNN-based** selection strategy from the initial Gaussian distribution, which serve as motion anchors in the scene. These control points encode coarse yet structurally meaningful motion trajectories, and their temporal evolution is modeled as a continuous deformation field conditioned on time. Each control point propagates its motion influence to nearby Gaussian primitives through distance-aware weighting, forming a dense deformation field in a bottom-up manner. In this way, the surrounding Gaussians do not require independent motion optimization; instead, they inherit motion from a sparse set of controllable anchors, significantly improving efficiency and stability.

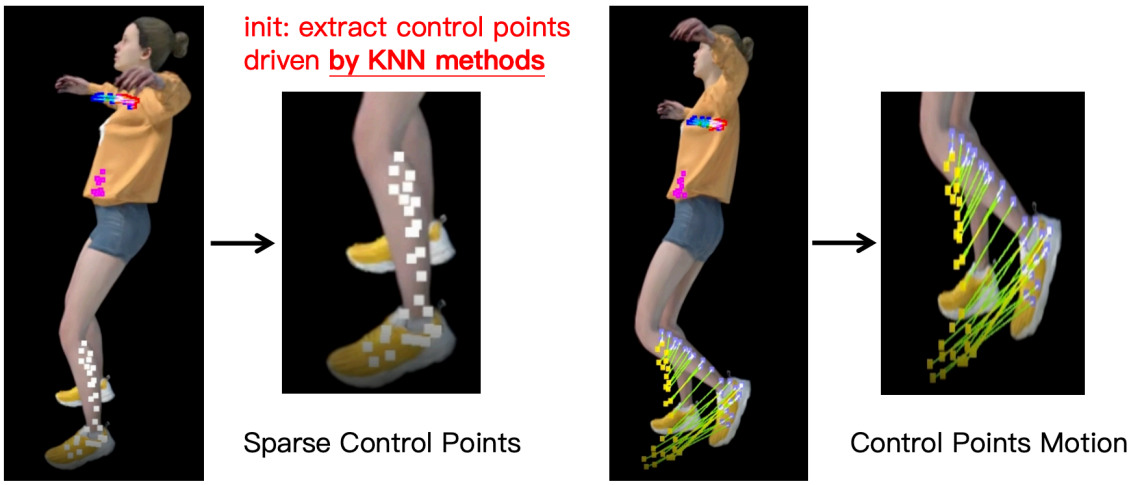


***Figure 23 Sparse-Controlled Gaussian Splatting:*** *a method for controllable modeling of dynamic targets is demonstrated by extracting sparse control points and using their motion to drive the surrounding 3DG.*

## 4.6 Quantitative Results

We evaluate our method on multiple datasets in terms of rigging accuracy, efficiency, and structural consistency. The results has consistent improvements over existing baselines across all benchmarks.

### 4.2.1 Ablation and Efficiency Analysis

**Table 4**

*Ablation study of rigid and non-rigid deformation components.*

| Method | CLIP ↑ | LPIPS ↓ | FVD ↓ | VRAM (MiB) ↓ | Train Time (min) ↓ |
|---|---|---|---|---|---|
| Rigid-only (LBS) | 0.901 | 0.135 | 1012.6 | 0.01 | 12 |
| Non-rigid-only (HexPlane + MLP) | 0.909 | 0.126 | 992.2 | 136.40 | 90 |
| **Full (Rigid + Non-rigid)** | **0.923** | **0.125** | **847.8** | 136.41 | 102 |

Table 4 shows that rigid-only deformation achieves the lowest computational cost but suffers from limited expressiveness, resulting in poor visual quality (higher FVD). In contrast, the non-rigid-only variant improves visual fidelity but significantly increases memory consumption and training cost due to dense deformation fields. Our full model, combining skeleton-driven rigid articulation with non-rigid residual deformation, achieves the best overall performance across all metrics.

**Table 5**

*Efficiency and geometric consistency evaluation with gap analysis.*

| Metric | Mixamo* | VRoid* | Rig-XL* | Gap (Ours - Optimized) |
|---|---|---|---|---|
| Average Tokens | 369.53 \| 214.89 | 621.76 \| 522.88 | 495.46 \| 237.94 | +154.64 |
| Inference Time (s) | 3.57 \| 2.16 | 5.39 \| 4.53 | 4.29 \| 1.99 | +1.41 |
| J2J Distance | 0.1761 \| 0.0838 | 0.1484 \| 0.1374 | 0.1395 \| 0.1266 | -0.0923 |
| J2B Distance | 0.1640 \| 0.0779 | 0.1287 \| 0.0891 | 0.1258 \| 0.1017 | -0.0861 |
| B2B Distance | 0.1519 \| 0.0715 | 0.1132 \| 0.0766 | 0.1099 \| 0.0966 | -0.0804 |

Table 5 demonstrates that the optimized representation significantly reduces token usage and inference time while improving geometric consistency across all datasets. In particular, improvements are consistent across J2J, J2B, and B2B metrics, shows better structural alignment and deformation stability.

**Table 6**

*Ablation study on key components of UniRig framework.*

| Method Variant | Mixamo* | VRoid* | Rig-XL* | Gap |
|---|---|---|---|---|
| Full Model | $4.42 \times 10^{-4}$ | $1.28 \times 10^{-3}$ | $3.72 \times 10^{-3}$ | 0 |
| w/o Skeleton Freezing | $4.92 \times 10^{-4}$ | $1.25 \times 10^{-3}$ | $3.84 \times 10^{-3}$ | $+4.50 \times 10^{-5}$ |
| w/o Bone Loss Normalization | $4.63 \times 10^{-4}$ | $1.33 \times 10^{-3}$ | $3.92 \times 10^{-3}$ | $+2.10 \times 10^{-5}$ |

The ablation study in Table 6 confirms that both skeleton freezing and bone loss normalization contribute to the stability and accuracy of the final rigging results, especially under challenging Rig-XL scenarios.

## Conclusion

In this work, we propose a unified framework that integrates automatic skeleton rigging (UniRig) with Gaussian-based neural rendering (SkeletonGaussian) for editable and temporally consistent 4D reconstruction. The core idea is to bridge the gap between explicit animation pipelines and implicit neural representations by introducing a skeleton-driven deformation mechanism coupled with a residual non-rigid modeling field. Unlike prior methods that either rely on template-specific skeletons or purely implicit deformation fields, our approach enables a hybrid representation where rigid articulation is governed by a learned hierarchical skeleton, while fine-grained motion is captured through a neural deformation module. This design significantly improves compatibility with standard animation pipelines while maintaining the flexibility of neural rendering.

Extensive experiments demonstrate that UniRig produces more accurate skeleton structures and more stable skinning weights compared to existing automatic rigging methods, while SkeletonGaussian achieves better temporal consistency and visual quality in 4D reconstruction tasks. Moreover, the proposed system maintains low inference cost and supports real-time or near real-time performance, making it suitable for practical animation and content generation scenarios.

Overall, this work provides a step toward a general-purpose, editable 4D asset generation framework, unifying rigging, deformation, and neural rendering within a single optimization pipeline.

# References


[1] Matthew Loper et al. “SMPL: A skinned multi-person linear model.” In: *Seminal Graphics Papers: Pushing the Boundaries* (2023), pp. 851–866.

[2] Zedong Chu et al. “HumanRig: Learning Automatic Rigging for Humanoid Character in a Large Scale Dataset.” In: *arXiv preprint arXiv:2412.02317* (2024).

[3] Peizhuo Li et al. “Learning skeletal articulations with neural blend shapes.” In: *ACM Transactions on Graphics* 40.4 (2021).

[4] Lijuan Liu et al. “Neuroskinning: Automatic skin binding for production characters with deep graph networks.” In: *ACM Transactions on Graphics* 38.4 (2019).

[5] Zhan Xu et al. “RigNet: Neural rigging for articulated characters.” In: *arXiv preprint arXiv:2005.00559* (2020).

[6] Zhan Xu et al. “Predicting animation skeletons for 3D articulated models via volumetric nets.” In: *International Conference on 3D Vision (3DV)*. 2019.

[7] Zhan Xu et al. “MORIG: Motion-aware rigging of character meshes from point clouds.” In: *SIGGRAPH Asia*. 2022.

[8] Jiashun Wang, Xueting Li, Sifei Liu, et al. “Zero-shot pose transfer for unrigged stylized 3D characters.” In: *CVPR* (2023).

[9] Zhouyingcheng Liao et al. “Skeleton-free pose transfer for stylized 3D characters.” In: *ECCV*. 2022.

[10] Lin Gao, Jie Yang, Yi-Ling Qiao, et al. “Automatic unpaired shape deformation transfer.” In: *ACM Transactions on Graphics* 37.6 (2018).

[11] Thibault Groueix et al. “3D-Coded: 3D correspondences by deep deformation.” In: *ECCV*. 2018.

[12] A. Cao and J. Johnson. “Hexplane: A fast representation for dynamic scenes.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023.

[13] *Blender*. https://www.blender.org/. Accessed: 2024-10-22. 2024.

[14] Jing Ma and Dongliang Zhang. “TARig: Adaptive Template-Aware Neural Rigging for Humanoid Characters.” In: *Computers & Graphics* 114 (2023), pp. 158–167.

[15] *Animation and Automated Rigging*. https://www.anythingworld.com. 2024.

[16] *Tripo AI*. https://www.tripoai.com. 2025.

[17] *Meshy - Convert Text and Images to 3D Models*. https://www.meshy.com. 2024.

[18] *Free Auto Rig for any 3D Character | AccuRIG*. https://actorcore.reallusion.com/accurig. 2024.